

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

July 2022

Course Code: **MEC101**

Course: **BASIC MECHANICAL ENGINEERING**

Question Paper No. : **U0121**

Duration: **3 hours**

Max. Marks: **100**

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. Compare the operation of making hole in lathe and drilling machine.
2. What happened if the tool overhang in the tool holder is more while machining in the shaper?
3. State the principle of rolling process.
4. What is the use of gate cutter in the moulding process?
5. How the temperature of the flame in gas welding process can be controlled?
6. List any four properties of refrigerant.
7. Why air cooling system is not suitable for medium and large capacity engines?
8. State the function of air preheater and economizer in thermal power plant.
9. What are the salient features of Benson boiler?

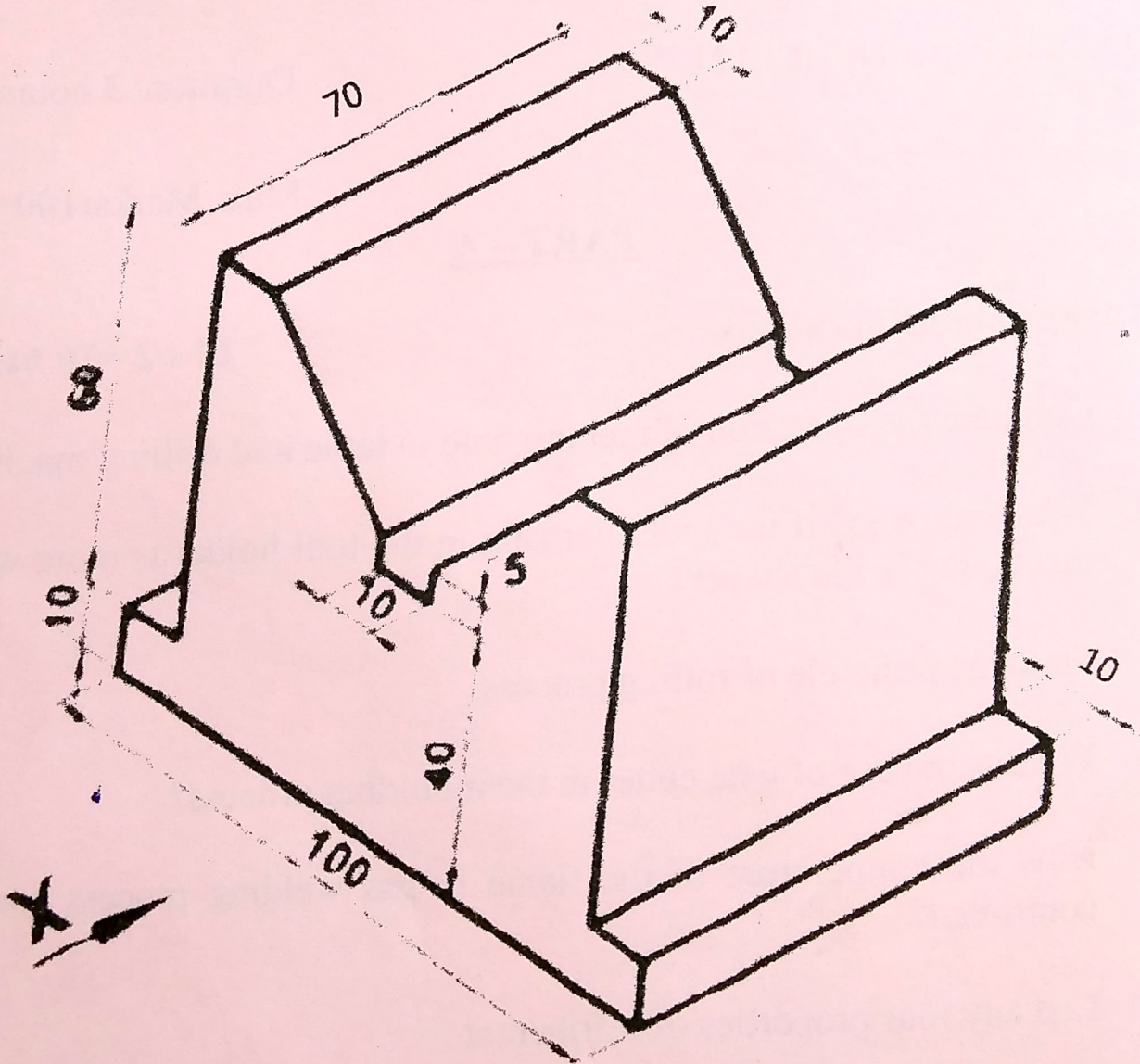
10. Compare the working of impulse and reaction turbine.

PART - B

Answer all the questions

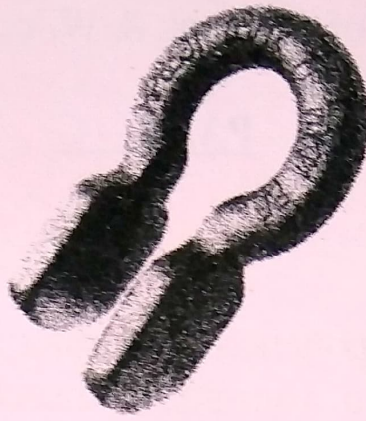
4 x 15 = 60 Marks

11. Identify the various operations needed to make the component given below from a cube of size 100 mm.



(OR)

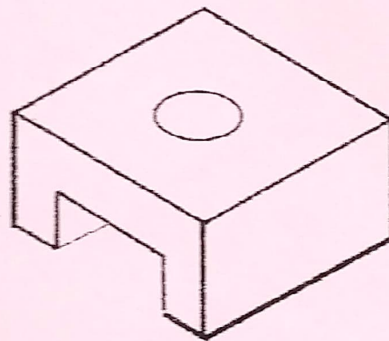
12. Identify the suitable process to fabricate the component given below from a straight rod of circular cross section and explain the same with necessary diagrams.



13. Explain the manual metal arc welding process with need sketch.

(OR)

14. Elaborately discuss the process of green sand moulding for the component shown below:



15. Discuss the vapor compression refrigeration system and explain the functions of each component.

(OR)

16. How supply of fuel in petrol engine is different from diesel engine. Explain the fuel supply system in petrol engine with neat sketch.

17. Draw the neat sketch for combined cycle power plant and explain its working.

(OR)

18. Explain the working of Babcock & Wilcox Boiler with neat diagram.

PART – C

Answer the following

1 x 20 =20 Marks

19. List the major component of an internal combustion engine and explain the working of four stroke diesel engine with neat diagram. Also explain the fabrication method for any two components in the same engine.
